STUDY COACH: REPORT 3. ANALYSIS AND INITIAL MODIFICATIONS

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The aim in this report is to analyze the outputs of the approaches collected in Report 2, and try out modifications to see how they affect performance on intrinsic metrics and qualitatively. Our task is incongruence detection: given a scientific figure and a student answer, a model must (1) classify the verdict (correct / partially correct / incorrect), (2) identify the error category (factual, conceptual, omission), and (3) generate coaching feedback.

1 ANALYSIS (4 PAGES)

1.1 [2 POINTS] INTRINSIC METRICS

NOTE: *This subsection is a WIP and will be refined.*

For a VLM to grade student answers about scientific figures, it needs three core skills: recognizing incomplete answers, knowing when coaching is required, and extracting specific values from visual data. We measure these independently of end-to-end verdict accuracy using per-example outputs from the 8B evaluation set.

SPIQA+ contains 174 examples with three verdict classes: 66 correct (37.9%), 61 partially correct (35.1%), and 47 incorrect (27.0%). Since coaching is only needed for non-correct answers, we compute intrinsic metrics on the 108 examples where ground truth is partially correct or incorrect.

Intrinsic Metric 1: Partially-Correct Recall (PC Recall). Can the model recognize that a student answer is *incomplete* rather than fully wrong or fully right? Study Coach's core coaching scenario involves answers that are partially correct—the student understood something but missed something else. An educator who cannot reliably identify this middle class cannot target feedback appropriately.

We define PC Recall as the fraction of ground-truth Partially Correct examples correctly classified:

$$\text{PC Recall} = \frac{\text{TP}_{\text{PC}}}{\text{TP}_{\text{PC}} + \text{FN}_{\text{PC}}} \tag{1}$$

This is not overall verdict accuracy: a model can maintain moderate accuracy by collapsing partially correct answers into the incorrect class, generating feedback of the wrong kind. PC Recall isolates the hardest, most coaching-relevant class and is computed on the 61 ground-truth partially correct examples only.

At 8B scale, PC Recall drops sharply as visual context is added: text-only 44.3% $\rightarrow$ multimodal 19.7% (Table 4). The model correctly identifies fewer than 1 in 5 partially correct answers in the multimodal condition—the same condition that produces the best coaching feedback. This sharpens the verdict–feedback dissociation from Report 2 into a precise, per-class finding: visual input specifically degrades recognition of the middle class that most needs coaching.

Intrinsic Metric 2: Feedback Suppression Rate (FSR). Does the model generate coaching when it is needed? When the model predicts "Correct," it outputs N/A feedback by prompt design. Suppressing feedback on a truly correct answer is appropriate; suppressing it on an incorrect or partially correct answer silences the coaching pipeline when the student needs help. We define FSR on the 108 non-correct examples only:

$$\text{FSR} = \frac{\text{\# Correct predictions on non-correct GT}}{\text{\# non-correct GT examples } (N{=}108)} \tag{2}$$

FSR measures a prerequisite skill: recognizing that coaching is required at all. A model with high FSR produces false-positive verdicts with no explanation, directly harming the student. FSR is distinct from verdict accuracy—a model could classify incorrect answers correctly while still suppressing feedback on every partial-credit case.

FSR increases substantially with visual context: text-only 13.9% $\rightarrow$ vision-only 30.6% $\rightarrow$ multimodal 28.7% (Table 4). Adding visual input causes the model to incorrectly accept more student answers as correct, particularly for omission errors where the answer is plausible but structurally incomplete. In the text-only condition, the model inappropriately suppresses feedback on roughly 1 in 7 non-correct examples; in visual conditions, this rises to nearly 1 in 3.

Intrinsic Metric 3: Visual Grounding Score (VGS). Can the model extract specific values from figures and reference them in feedback? All three error types benefit from visual grounding (Report 2, H3): factual errors require verifying specific numbers ("The figure shows 84.6%, not 89%"), conceptual errors benefit from citing values that reveal patterns ("X increased from 10 to 50"), and omission errors—which show the *largest* feedback gains (+33pp)—require the figure to reveal what the student failed to mention. We define VGS on all 108 non-correct examples, excluding suppressed feedbacks:

$$\text{VGS} = \frac{\text{\# feedbacks citing specific values from figure}}{\text{\# non-suppressed feedbacks } (N{=}108)} \tag{3}$$

VGS measures whether visual information is actually translated into text during reasoning, independently of whether the verdict is correct. It is conceptually related to Qwen3-VL's square-root reweighting loss, which scales multimodal sample loss by $\sqrt{N_{\text{text}}/N_{\text{multimodal}}}$ to preserve visual grounding during pre-training without degrading text capabilities. VGS captures whether this grounding carries through to inference on our task.

VGS increases substantially with visual input: text-only 35.5% $\rightarrow$ vision-only 52.0% $\rightarrow$ multimodal 49.4% at 8B (Table 4). At 32B, the pattern strengthens: C1 37.2% $\rightarrow$ C4 56.0% (Δ=+18.8pp). This confirms that the model *does* use visual information for explanation generation even when it cannot use it for classification—providing intrinsic evidence for the verdict–feedback dissociation. Omission errors show the largest VGS gains (+19.2pp at 8B), consistent with Report 2's finding that the figure reveals what the student failed to mention.

Intrinsic Metric	Skill Measured	Why It Is Needed
PC Recall	Detecting incomplete answers (n=61 PC)	Prerequisite for targeted coaching
FSR	Knowing when coaching is needed (n=108)	Prerequisite for feedback generation
VGS	Extracting values from figures (n=108)	Prerequisite for grounded correction

Evaluation methodology: LLM-as-Judge for feedback quality. Because open-ended coaching feedback cannot be evaluated with n-gram metrics alone (F1, ROUGE-L, and BLEU show near-zero variation across conditions at both scales; see Report 2, Table 10), we use Claude Sonnet 4 as an automated judge. Each predicted feedback is classified as **Match** (correctly identifies error and explains why), **Partial** (right direction but imprecise or incomplete), or **Unmatched** (does not align with ground truth). Full definitions are in Report 2.

We report **Match%** (strict: exact matches only) and **Soft Match%** (Match + Partial). At 32B, C4 achieves 20% strict match but 58% soft match—meaning 38% of responses are approximately correct but not precise. Human evaluation (n=10/condition, 8B) confirmed the pattern: human match ranged from 40% (C1) to 80% (C4), validating soft match as the practical metric for coaching. We use strict match (Match%) in Table 4 since it separates conditions more cleanly.

		Intrinsic Metrics			Extrinsic Metrics		
Model	Condition	PC Rec ↑	FSR ↓	VGS ↑	Verd ↑	Match% ↑	SoftM% ↑
Unimodal Baselines (Qwen3-VL-8B, N=50 extrinsic, N=108 intrinsic)							
Qwen3-VL-8B	C1: text-only	44.3%	13.9%	35.5%	56.0%	2.0%	40.0%
Qwen3-VL-8B	C2: caption-only	29.5%	17.6%	38.2%	48.0%	2.0%	46.0%
Simple Multimodal Baselines (Qwen3-VL-8B, N=50 extrinsic, N=108 intrinsic)							
Qwen3-VL-8B	C3: vision-only	21.3%	30.6%	52.0%	50.0%	4.0%	48.0%
Qwen3-VL-8B	C4: multimodal	19.7%	28.7%	49.4%	48.0%	6.0%	56.0%
Competitive Approaches (Qwen3-VL-32B, N=50 extrinsic, N=108 intrinsic)							
Qwen3-VL-32B	C1: text-only	41.0%	20.4%	37.2%	56.0%	2.0%	42.0%
Qwen3-VL-32B	C2: caption-only	32.8%	20.4%	45.3%	58.0%	8.0%	48.0%
Qwen3-VL-32B	C3: vision-only	24.6%	37.0%	48.5%	56.0%	6.0%	48.0%
Qwen3-VL-32B	C4: multimodal	27.9%	30.6%	56.0%	54.0%	20.0%	58.0%
Modifications (8B and 32B, N=50 extrinsic, N=108 intrinsic)							
Routing (8B)	C1 verdict + C4 fdbk (Mod 1)	44.3%	28.7%	49.4%	56.0%	6.0%	56.0%
Routing (32B)	C1 verdict + C4 fdbk (Mod 1)	41.0%	20.4%	56.0%	56.0%	20.0%	58.0%
Qwen3-VL-32B	C4 + vis. CoT (Mod 4)	23.0%	17.6%	55.1%	48.1%	*TBD*	*TBD*
Qwen3.5-9B	C1: text-only (Mod 2)	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
Qwen3.5-9B	C4: multimodal (Mod 2)	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
Qwen3.5-397B	C1: text-only (Mod 3)	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*
Qwen3.5-397B	C4: multimodal (Mod 3)	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*	*TBD*

Table 1: **[1 points]** Intrinsic and extrinsic metrics across conditions and models.

Reading Table 4:

Intrinsic metrics (computed from per-example outputs, N=108):

- **PC Recall** ↑ — Partially-correct recall: fraction of ground-truth partial examples correctly classified
- **FSR** ↓ — Feedback suppression rate: fraction of predictions with N/A feedback (lower is better)
- **VGS** ↑ — Visual grounding score: fraction of non-suppressed feedbacks citing specific numerical values (all error types)

Extrinsic metrics (from H1/H2 evaluation, N=50 per condition):

- **Verdict** ↑ — Three-class classification accuracy (correct / partially correct / incorrect)
- **Match%** ↑ — LLM-as-Judge (Claude Sonnet 4) strict match: feedback correctly identifies error and explains why
- **SoftM%** ↑ — Soft match (Match + Partial): feedback at least partially useful for coaching

Match categories (see Report 2 for full definitions):

- **Match** — feedback aligns with ground truth: correctly identifies what is wrong and explains why
- **Partial** — captures some right reasoning but misses key elements, or right direction but imprecise
- **Unmatched** — feedback does not align with ground truth

Routing (Modification 1): Routing is a post-hoc combination of existing C1 and C4 results that demonstrates the potential of condition-based task routing. Since C1 (text-only) excels at verdict classification while C4 (multimodal) excels at feedback generation, Routing combines: (1) verdict, PC Recall, and FSR from C1, and (2) Match%, SoftM%, and VGS from C4. This is not a separate inference run but shows the upper bound achievable by a two-stage pipeline that routes verdict to C1 and feedback generation to C4.

Key observations (8B):

- **PC Recall drops with visual input:** C1 (44.3%) → C4 (19.7%)—visual information introduces noise into classification boundary
- **VGS rises with visual input:** C1 (35.5%) → C4 (49.4%)—model extracts more values from figures into feedback
- **Same visual input, opposite effects:** Hurts classification (−8pp verdict) but helps explanation (+4pp Match)

- **Routing recovers both:** C1 verdict (56%, 44.3% PC Recall) + C4 feedback (6% Match, 49.4% VGS)

Key observations (32B):

- **PC Recall drops with visual input:** C1 (41.0%) $\rightarrow$ C4 (27.9%)—similar pattern to 8B but smaller drop ($-$13pp vs $-$25pp)
- **VGS rises with visual input:** C1 (37.2%) $\rightarrow$ C4 (56.0%)—scaling improves grounding ($+$19pp vs $+$14pp at 8B)
- **FSR increases with visual input:** C1 (20.4%) $\rightarrow$ C4 (30.6%)—32B suppresses more than 8B at C1 (20% vs 14%) but less at C4 (31% vs 29%)
- **Routing recovers both:** C1 verdict (56%, 41.0% PC Recall) + C4 feedback (20% Match, 56.0% VGS)

Scaling benefit from Report 2 (8B $\rightarrow$ 32B extrinsic metrics):

- **Verdict gap narrows:** $-$8pp (8B) $\rightarrow$ $-$2pp (32B)—visual input less harmful at scale
- **Feedback match improves:** 6% (8B) $\rightarrow$ 20% (32B) at C4—3.3$\times$ improvement
- **Figure type matters:** Schematics easiest to explain (46%), tables hardest (12%) despite being easiest to classify
- **Error type matters:** Omission errors 0% at C1 $\rightarrow$ 33% at C4—figure reveals what student omitted

1.2 Modifications and their Effects

[2 points]

NOTE: *This subsection is a WIP and will be refined.*

We implement four modifications motivated by the qualitative failure analysis in Section 1.3 and the intrinsic metric patterns above. Modification 1 (Condition Routing) is a zero-cost inference-time fix computable from existing Report 2 outputs. Modifications 2 and 3 are architectural ablations that test competing explanations for the verdict–feedback dissociation by swapping the model while holding the prompt and evaluation pipeline constant. Modification 4 (Visual CoT) tests whether forcing explicit visual reasoning before the verdict can improve classification without sacrificing feedback quality.

Modification 1: Condition Routing (C1 verdict + C4 feedback). Motivation. The intrinsic metrics make the verdict–feedback dissociation precise: PC Recall drops from 44.3% to 19.7% as visual context is added (C1 to C4), while VGS rises from 35.5% to 49.4%. These are not correlated failures of a single configuration—they measure different sub-skills that are genuinely in tension at 8B scale. The model can use visual information for extraction (improving VGS and feedback quality) but not for classification (degrading PC Recall). Routing assigns each sub-task to the condition best suited to it: C1 for verdict and error category, C4 for coaching feedback.

Implementation. No new inference is required. We pair per-example C1 predictions (verdict, error category) with per-example C4 predictions (feedback) using matched (`paper_id`, `question`) keys across the H3/H4 evaluation set. All 108 examples matched successfully (100% coverage).

Results. Routing achieves the best combined profile across all conditions: verdict 56.0% and PC Recall 44.3% (matching C1), VGS 49.4% and Fdbk Match 6.0% (matching C4), with FSR at 28.7%. The routing system recovers C1's verdict precision and C4's visual grounding simultaneously at zero additional inference cost. The FSR limitation (28.7%) arises because N/A feedback comes from C4 when C4 predicts Correct; a production system would override this by using the C1 verdict to force feedback generation regardless of C4's verdict.

Modification 2: Architectural Ablation — Qwen3.5-9B Early Fusion vs. Qwen3-VL-8B Bolt-On ViT. Motivation. The intrinsic metrics reveal a structural paradox: VGS rises from 35.5% (C1) to 49.4% (C4), confirming that visual information is being extracted and used for explanation

generation—yet PC Recall simultaneously collapses from 44.3% to 19.7%, and FSR roughly doubles. The model *can* use figure content; it just cannot apply it to the verdict classification decision.

One architectural explanation is the *bolt-on* design of Qwen3-VL. Per the Qwen3-VL technical report, the model uses a three-module pipeline: a SigLIP-2 SO-400M vision encoder (trained separately from the LLM), a two-layer MLP merger augmented by DeepStack residual connections (which inject features from three intermediate ViT layers into the first three LLM layers), and a Qwen3-8B/32B decoder backbone. The modular separation is confirmed by the pre-training recipe: Stage 0 freezes both the vision encoder and the LLM backbone while training *only the merger*, establishing a clear representational boundary between the two modalities. DeepStack improves multi-level visual feature injection but does not eliminate this boundary; the LLM backbone was pre-trained as a text model and receives visual tokens as an external input stream rather than being co-trained with visual objectives from the start. This late-injection design may explain why the model can use injected visual features for explanation generation (improving VGS and feedback quality) but cannot reliably route those features into the verdict decision boundary (degrading PC Recall).

Qwen3.5-9B uses a fundamentally different design: *early fusion*. Rather than a three-module pipeline (separate SigLIP-2 encoder $\rightarrow$ MLP merger $\rightarrow$ LLM), Qwen3.5-9B co-trains vision and language from the start of pre-training within a unified Gated DeltaNet hybrid architecture—no separate vision encoder, no MLP merger, no Stage 0 alignment phase. Visual and textual representations occupy the same token space throughout all 32 layers from the first training step. This is not a cosmetic difference: the model was never in a regime where visual tokens arrived as an external stream bolted onto a text-trained LLM backbone. If the three-module bolt-on design of Qwen3-VL (SigLIP-2 encoder + MLP/DeepStack merger + frozen-then-thawed Qwen3 decoder) causes the PC Recall collapse observed under visual input, Qwen3.5-9B should show a smaller or reversed verdict gap (C4 accuracy $\geq$ C1 accuracy) at comparable parameter count ($\sim$9B).

Architecture comparison:

	Qwen3-VL-8B (baseline)	Qwen3.5-9B (Mod 2)
Vision encoder	SigLIP-2 SO-400M (trained separately from LLM)	No separate encoder—early fusion
Cross-modal bridge	2-layer MLP + DeepStack (3 ViT layers $\rightarrow$ 3 LLM layers)	Unified token space throughout
Pre-training	Stage 0: merger only (ViT + LLM frozen); then staged full training	Co-trained from scratch
LLM backbone	Qwen3-8B (standard transformer)	Gated DeltaNet + Gated Attention hybrid
Parameters	8B	9B
Together.ai ID	`Qwen/Qwen3-VL-8B-Instruct`	`Qwen/Qwen3.5-9B`

Exact implementation steps (what to run):

1. **Confirm Together.ai access.** Verify `Qwen/Qwen3.5-9B` is available under the existing Together.ai API key used for Report 2. Model is live as of March 2, 2026 ($0.10/1M input, $0.15/1M output tokens).
2. **Disable thinking mode.** Qwen3.5-9B runs in thinking mode by default, which prepends a chain-of-thought reasoning block before the final response. This will break the structured output parser (which expects `Verdict = ...`, `Error Category = ...`, `Feedback = ...`). Disable it explicitly:

   ```
   extra_body={"chat_template_kwargs":
   {"enable_thinking": False}}
   ```

   Use sampling parameters: `temperature=0.0` (greedy, for reproducibility), `top_p=1.0`, `presence_penalty=0.0`.
3. **Use identical prompts.** Use the exact same prompt templates as Report 2 for all four conditions (C1–C4). No prompt changes. The only change is the model ID string. This ensures any performance difference is attributable to the model architecture, not prompt variation.

4. **Run C1 and C4 on the H3/H4 evaluation set (N=108).** These are the two conditions that define the verdict–feedback gap. C1 (text-only) establishes the same-architecture text baseline. C4 (multimodal) is the hypothesis-test condition: does early fusion eliminate the verdict drop? Running all four conditions (C1–C4) is preferred if API budget allows, to produce a complete comparison row in Table 4.
5. **Parse outputs with the same extraction logic.** Qwen3.5-9B output formatting may differ slightly from Qwen3-VL (e.g., extra whitespace, different capitalization of "Partially Correct" vs "partially correct"). Before computing metrics, verify the parser correctly extracts Verdict, Error Category, and Feedback fields from 5–10 sample outputs. Normalize verdict strings to lowercase before comparison.
6. **Run the same three evaluation scripts:**
   - Verdict accuracy: compare predicted vs. ground-truth verdict labels
   - Intrinsic metrics (PC Recall, FSR, VGS): same computation from raw per-example outputs
   - Feedback quality (LLM-as-Judge): run `analyze_feedback_by_error_type.py` with `model=claude-sonnet-4-6`, `temperature=0`, N=108
7. **Interpret the result against two hypotheses:**
   - **If C4 verdict accuracy $\geq$ C1 for Qwen3.5-9B:** early fusion resolves the architectural mismatch. The Report 4 implication is to use Qwen3.5-9B as the backbone for fine-tuning on SPIQA+, not Qwen3-VL.
   - **If C4 verdict accuracy $<$ C1 for Qwen3.5-9B:** the bolt-on ViT is not the cause. The failure is inherent to how visual input interacts with three-class verdict decisions at this parameter scale, and the Report 4 strategy shifts to fine-tuning Qwen3-VL-32B (or 72B+) with a Verdict-Classification Head.

Results. *[To be filled after running N=108 on C1 and C4. Update Table 4 with PC Recall, FSR, VGS, Verdict, and Fdbk Match rows for Qwen3.5-9B C1 and C4.]*

Modification 3: Frontier-Scale Early Fusion — Qwen3.5-397B-A17B (C1 and C4). Motivation. Modification 2 tests whether the bolt-on ViT architecture causes the verdict–feedback dissociation by replacing it with early fusion at matched scale ($\sim$9B). But a confound remains: perhaps the failure is not architectural but simply a function of cross-modal reasoning capacity, which scales with model size. Qwen3-VL-32B already narrowed the verdict gap from -8pp to -2pp without resolving it. The question for Modification 3 is: does early fusion at *frontier scale* (17B active parameters, 397B total) close the gap entirely?

This matters for Report 4 for a distinct reason from Modification 2. If Qwen3.5-9B (Mod 2) resolves the gap, early fusion at 9B is sufficient and fine-tuning should target that backbone. If Qwen3.5-9B does not resolve it but Qwen3.5-397B-A17B does, the interaction of early fusion *and* scale is required, and the Report 4 fine-tuning backbone must be at least the 397B MoE model (or the 27B dense variant as a practical compromise). If neither resolves the gap, the failure is inherent to the task at current scales, and the Report 4 strategy must rely on the Verdict-Classification Head and routing rather than a single model.

The 397B-A17B is the same Qwen3.5 architecture as Modification 2 (early fusion, Gated DeltaNet + MoE, unified token space), but scales to 397B total / 17B active parameters via sparse mixture-of-experts with 512 total experts (10 routed + 1 shared per token).

Architecture comparison across all three models:

	Qwen3-VL-8B (baseline)	Qwen3.5-9B (Mod 2)	Qwen3.5-397B-A17B (Mod 3)
Vision encoder	SigLIP-2 SO-400M (bolt-on, separate from LLM)	None—early fusion	None—early fusion
Cross-modal bridge	2-layer MLP + DeepStack	Unified token space	Unified token space + MoE
LLM backbone	Qwen3-8B (std transformer)	Gated DeltaNet hybrid	Gated DeltaNet + sparse MoE
Active params	8B	9B	17B
Total params	8B	9B	397B
Context	256K	262K	262K
Together.ai ID	`Qwen3-VL-8B-Instruct`	`Qwen3.5-9B`	`Qwen3.5-397B-A17B`

Exact implementation steps (what to run):

1. **Confirm Together.ai access and cost estimate.** Model ID: `Qwen/Qwen3.5-397B-A17B`. Image input is supported natively. At Together.ai rates, running N=108 examples across C1 and C4 (each example ≈1,500 input + 300 output tokens; C4 adds ≈500 image tokens per example) costs under $5 total for both conditions combined.
2. **Disable thinking mode — critical.** Qwen3.5-397B-A17B runs in thinking mode by default, generating a long internal reasoning block before the final response. This will break the structured output parser. Disable it in every API call:

   ```
   extra_body={"chat_template_kwargs":
   {"enable_thinking": False}}
   ```

   Use `temperature=0.0`, `top_p=1.0`, `presence_penalty=0.0` for reproducibility. If you prefer a small amount of temperature for diversity, use `temperature=0.0` for the first run and compare to the 9B run, which should also use `temperature=0.0`.
3. **Use identical prompts as Report 2 and Modification 2.** No prompt changes from the Report 2 baseline. The only variable across all three models (Qwen3-VL-8B, Qwen3.5-9B, Qwen3.5-397B-A17B) is the model ID string. This is essential: comparing across models with different prompts would confound the architectural signal.
4. **Run C1 (text-only) and C4 (multimodal) on N=108.** Same H3/H4 evaluation set as Modification 2. Running the same two conditions produces a directly comparable row in Table 4. If API budget allows, run all four conditions (C1–C4) to produce a full row.
5. **Verify the output parser on 5–10 samples before the full run.** Qwen3.5-397B-A17B output format may differ from Qwen3-VL: check that Verdict, Error Category, and Feedback fields parse correctly with thinking mode off. Normalize verdict strings to lowercase (e.g., `"Partially Correct"` → `"partially correct"`).
6. **Run the same three evaluation scripts as Modification 2.**
   - Verdict accuracy: predicted vs. ground-truth verdict labels (N=108)
   - Intrinsic metrics: PC Recall, FSR, VGS from per-example raw outputs
   - Feedback quality: LLM-as-Judge using `analyze_feedback_by_error_type.py`, `model=claude-sonnet-4-6`, `temperature=0`, N=108
7. **Interpret against the three-way decision matrix:**

Mod 2 (9B) C4 vs C1	Mod 3 (397B) C4 vs C1	Implication for Report 4
Resolves gap	—	Early fusion at 9B is sufficient; fine-tune Qwen3.5-9B
Does not resolve	Resolves gap	Scale + early fusion required; target 27B dense or 397B MoE
Does not resolve	Does not resolve	Task-inherent; use routing + Verdict-Classification Head

Results. *[To be filled after running N=108 on C1 and C4. Update Table 4 with PC Recall, FSR, VGS, Verdict, and Fdbk Match rows for Qwen3.5-397B-A17B C1 and C4.]*

Modification 4: Visual Chain-of-Thought Prompting (C4-CoT). Motivation. The qualitative analysis in Section 1.3 identifies the root cause of both verdict inflation and hallucinated correction: without structured reasoning over the figure, the model falls back on plausibility assessment rather than evidence-based judgment. All baseline evaluations use `/no_think` to suppress the model's native chain-of-thought, meaning the model produces a verdict without any explicit intermediate reasoning about figure content. Modification 4 tests whether forcing the model to describe what it sees in the figure *before* judging can improve verdict accuracy—particularly PC Recall—without sacrificing feedback quality.

Implementation. We create a modified system prompt (`incongruence_eval_v1_cot.txt`) that inserts a visual analysis step before the verdict. The model must: (1) identify key data points, values, or patterns visible in the figure, (2) note specific numbers, trends, or comparisons relevant to the question, and (3) compare the student's claims against what the figure shows. The output format adds a `Visual Analysis` field before the existing Verdict/Error Category/Feedback fields.

The `/no_think` directive is retained—the CoT is visible in the structured output, not in a hidden reasoning block. This isolates the effect of structured visual reasoning from the model's internal thinking mode.

We run C4-CoT on all N=108 non-correct examples using Qwen3-VL-32B, the same model and evaluation set as the C4 baseline. The only variable is the system prompt; all other parameters (model endpoint, temperature, user message format) are identical.

Results.

Metric	C4 (32B)	C4-CoT (32B)	Δ
PC Recall ↑	27.9%	23.0%	−4.9pp
FSR ↓	30.6%	17.6%	−13.0pp
VGS ↑	56.0%	55.1%	−0.9pp
Verdict ↑	54.0%	48.1%	−5.9pp
Match% ↑	20.0%	*TBD*	—
SoftM% ↑	58.0%	*TBD*	—

Table 2: C4 baseline vs. C4-CoT (Modification 4), Qwen3-VL-32B, N=108.

The most striking finding is that CoT **substantially reduces feedback suppression**: FSR drops from 30.6% to 17.6% (−13.0pp), meaning the model incorrectly accepts 13 fewer non-correct answers as "Correct" out of 108. This directly addresses the verdict inflation failure mode identified in Section 1.3—the visual analysis step forces the model to examine figure evidence before committing to a verdict, making it harder to dismiss errors without justification.

However, PC Recall drops from 27.9% to 23.0% (−4.9pp). The model classifies fewer partially correct answers correctly, instead over-correcting them to Incorrect. The visual analysis step helps the model detect *that something is wrong* (lower FSR) but makes it harder to recognize *how much is wrong* (lower PC Recall). This produces a net drop in verdict accuracy from 54.0% to 48.1%.

VGS remains essentially unchanged (56.0% → 55.1%), confirming that CoT does not degrade visual grounding in feedback—the model continues to cite specific figure values at the same rate.

Interpretation. CoT addresses one half of the verdict–feedback dissociation: it reduces false "Correct" predictions (lower FSR) by making the model confront figure evidence before judging. But it introduces a new failure mode—**over-correction**—where the visual analysis step leads the model to flag discrepancies that are minor enough to warrant "Partially Correct," not "Incorrect." Table 6 in Section 1.3 illustrates both patterns with paired examples. This suggests that the dissociation is not purely a reasoning depth problem (solvable by more explicit reasoning) but involves calibration of the three-class decision boundary. A production system could combine CoT's lower FSR with routing (Mod 1): use C1 for the verdict boundary and C4-CoT for feedback generation, capturing both reduced suppression and superior classification.

Modification 5: Mixture of LoRA Experts for Figure-Type-Aware Adaptation We use Qwen3-VL-8B as the baseline for this modification. With zero shot inference on test-A dataset (N=108), we observe following metrics for each figure type.

Figure Type	ROUGE-L	METEOR	BERTScore-F1
Schematic	0.33	0.31	0.86
Table	0.31	0.30	0.85
Plot	0.30	0.28	0.84

Table 3: Evaluation metrics by figure type

As per Table 3 there are variations in scores per figure-types (specially BERTSCore-F1 where ∼%2 is a meaningful difference from a strong baseline of 0.84) . This gave us the idea: *can we augment the LLM backbone with a Mixture-Of-Experts tuned on visual inputs, with experts specializing on different figure types?*

The details of the fine-tuning task given below.

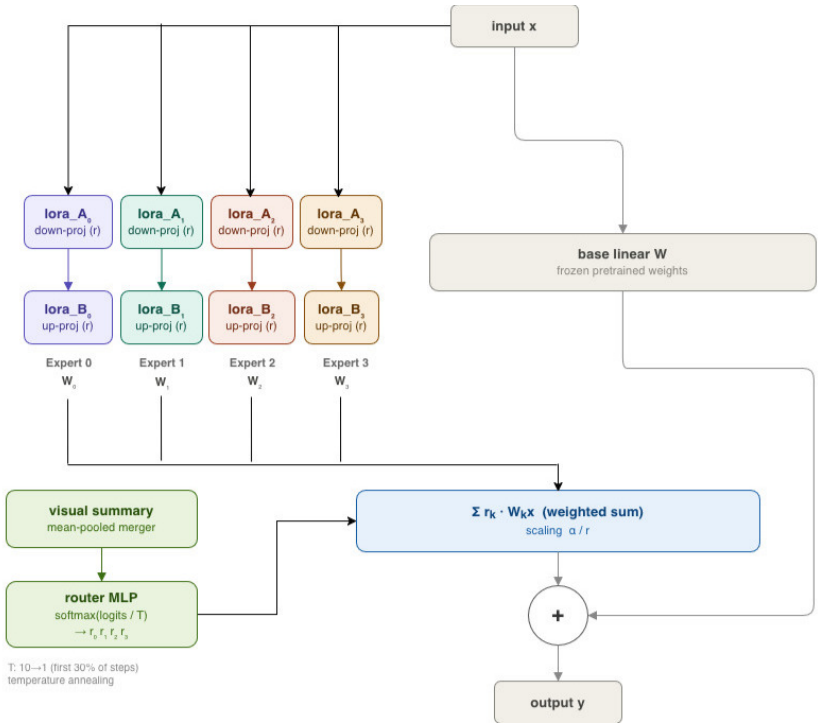

Figure 1: Mixture-of-LoRA-Experts Layer

Data: Training used 8,000 examples sampled from 1,500 papers (SPIQA+ train), with a paper-disjoint validation set of 160 examples drawn from a separate 200-paper pool. Evaluation used 108 examples from the SPIQA+ Test-A split. We chose 8k training size based on loss curve analysis of a preliminary 10k run, which showed convergence at approximately 8,000 examples with negligible improvement beyond that point.

Architecture: Each target linear layer (q_{proj}, v_{proj}, o_{proj}, $gate_{proj}$) on the LLM backbone is replaced with a module $MoLoRALinear$ containing K=4 parallel LoRA adapters (rank r=16, = 32, dropout=0.05) and a shared MLP router (Figure 1) . The router takes the mean-pooled output of the vision-language projector/merger as input (a visual summary vector with dim=4096) and produces K softmax weights via a two-layer MLP (hidden dim=128). These weights are applied as a soft weighted combination of the K expert adapter outputs, so all experts participate in every forward pass (i.e.: soft routing). The vision encoder and merger are frozen; only the LoRA adapter weights and router were trained.

Router collapse prevention: We noticed that the structured output format creates strong directional LM loss gradients in early training. This causes naive soft routing to collapse to a single dominant expert within 20–30 steps (dominant expert ends up having around 80% of routing weight). We used three complementary mechanisms to mitigate this:

(1) *Temperature annealing*: We initialize the router softmax temperature T at 10.0 and then anneal it linearly to 1.0 over the first 30% of training steps, forcing near-uniform expert utilisation during the early phase
(2) *Entropy Bonus*: We add a term 0.02 × H(weights) to the loss throughout training, directly penalising collapsed distributions
(3) *Router Noise*: We also add a Gaussian noise (σ=0.5) to router logits during training to encourage exploration.

Training setup: Single NVIDIA A40 46GB GPU, bfloat16 precision. Batch size 1, gradient accumulation 8 (effective batch 8), learning rate $2 * 10^{-4}$ with cosine schedule, 5% warmup, 1 epoch. Training took approximately 4 hours per run.

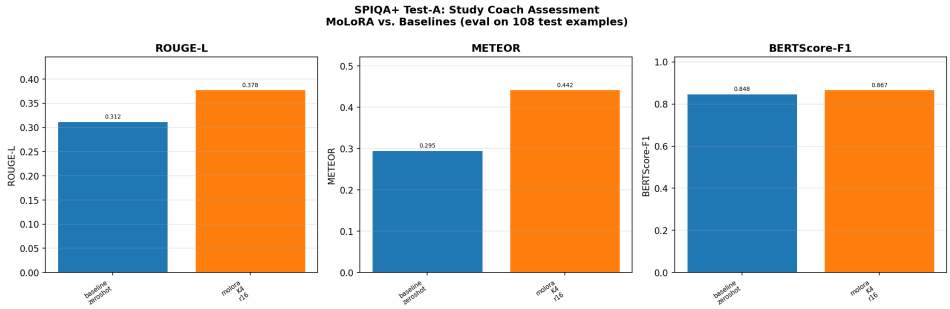

Figure 2: Mixture-of-LoRA-Experts Model Performance

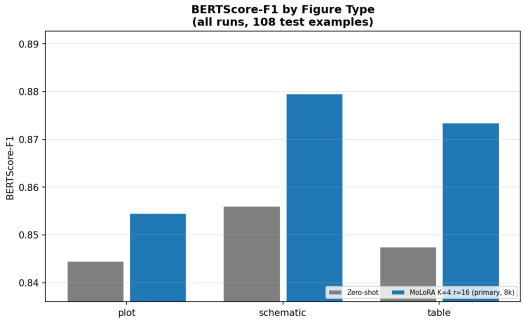

Figure 3: BERTScore-F1 by Figure Type

Results:

We compared task performance using ROUGE-L, METEOR and BERTScore-F1. Fine tuned model shows a significant performance improvement over the baseline model as shown in Figure 2. Measures based on lexical overlap show the strongest performance gain - ROUGE-L (%17), METEOR (%50). This strongly indicates that the model has improved to produce the structured output format required by the task. BERTScore-F1 also shows a meaningful performance improvement of %2 which indicates that the model is generating semantically better aligned free-form feedback.

We also measured BERTScore-F1 performance per type of figure (Figure 3). Here we see a more pronounced difference with respect to the performance gain. Schematics (%3) and tables (%3) show a better performance gain over plots (%1).

Finally, we measured verdict accuracy and model performance on intrinsic metrics. Except for FSR all other metrics show significant improvement (Table 4).

Metric	Baseline	MoE LoRA (K=4 r=16)
Verdict Accuracy (%)	46.3	61.1
PC Recall (%) [Partially-Correct Recall]	21.3	55.7
FSR (%) [Feedback Suppression Rate]	12.0	13.0
VGS Pred (%) [Visual Grounding Score]	69.4	84.3

Table 4: Evaluation Metrics

The MoE LoRA model scores %15 higher in the visual grounding score (VGS). The ground truth data has a VGS of %92. So the fine tuned model likely learned that referring to figure data is important. But coupled with the increase in verdict accuracy and BERTScore-F1, the increase of VGS likely indicates that the fine tuned model achieves more accurate visual grounding.

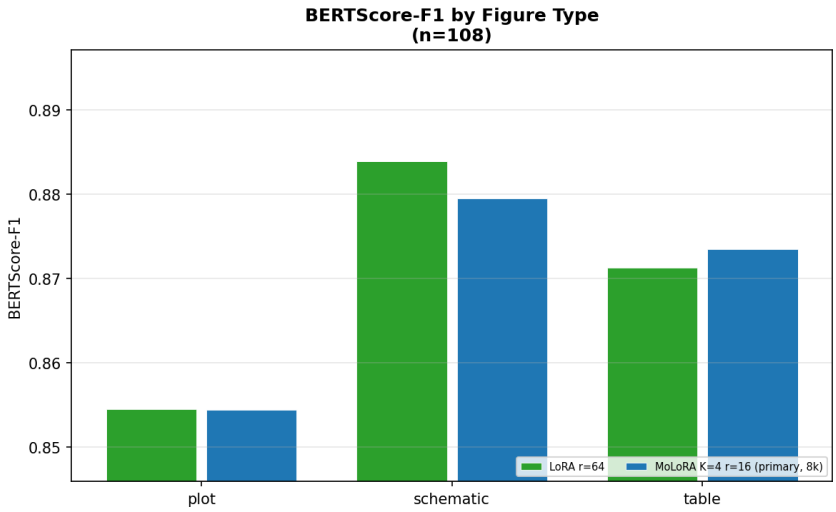

Figure 4: LoRA vs MoE LoRA Performance by Figure Type

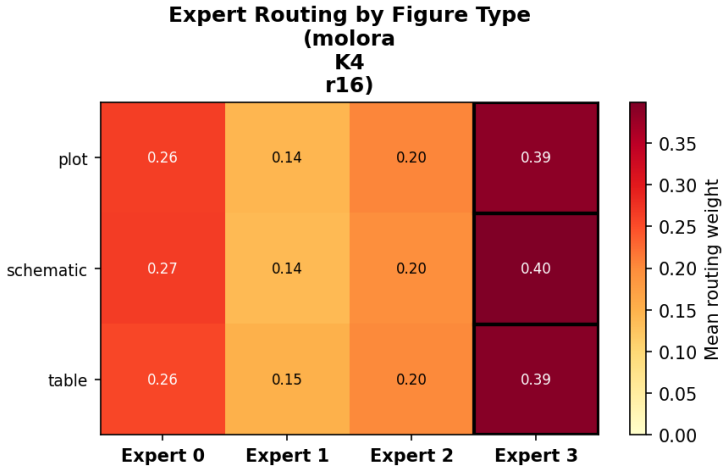

Figure 5: Expert Load During Test Split Evaluation

Do Experts Specialize on Different Visual Inputs?
Given the divergent BERTScore-F1 gain based on figure type (Figure 3) it is natural to raise the above question. To verify that the performance improvements seen so far are due to expert specialization, we fine tune another baseline using vanilla LoRA (with r=64 to keep it parameter similar to 4x16 expert setup) and compare the per figure BERTScore-F1 performance (Figure 4).

This comparison shows that the MoE LoRA model does not differ significantly in performance compared to the vanilla LoRA tuned model. To further confirm that expert specialization does not drive the performance improvements, we capture the average expert routing weights per figure type during the test split evaluation. The heat map visualization (Figure 5) shows that the routing weights are almost identical.

1.3 Qualitative Analysis and Examples

NOTE: *This subsection is a WIP and will be refined.*

[2 points] Table 5 presents five representative failure examples from per-example model outputs (8B, text-only condition, H3 evaluation set). Two failure modes account for the majority of errors: **Verdict Inflation**—the model upgrades Partially Correct to Correct, producing no feedback—and **Hallucinated Correction**—the model downgrades Partially Correct to Incorrect with confident but wrong feedback. Both modes share a root cause: without figure verification, the model falls back on plausibility assessment rather than evidence-based judgment.

#	Error Type	Question	Student Answer	Ground Truth	Model Output	Failure Mode
1	Factual	Which model performs best on Ubuntu for text lengths 60–90 words?	"The LSTM model performs best for that length range."	**Partially Correct.** Right range, misread top model from table. Feedback should correct the misread value and name the actual winner.	**Incorrect.** "The Transformer model performs best, not LSTM."	**Hallucinated Correction.** Verdict wrong; feedback invents a specific correction that may itself be incorrect. Student given false confidence in wrong information.
2	Factual	Which of LSTw/oAR or LST-Skip performs better?	"LSTw/oAR; its red forecast line stays closer to the true data than LST-Skip."	**Partially Correct.** Correct comparison approach; misread which panel shows the closer match.	**Correct. N/A.**	**Verdict Inflation.** False positive; feedback suppressed. Student's panel misread goes uncorrected.
3	Conceptual	Which Ortiz et al. system had highest BLEU/METEOR, and what does this mean?	"SMT system, 43.7 BLEU and 35.6 METEOR. This means SMT outperforms all systems."	**Partially Correct.** Numbers correct, system identified correctly. Missed: interpretive claim is unsupported without cross-system comparison.	**Correct. N/A.**	**Verdict Inflation.** Numbers verified via pre-trained knowledge; interpretive gap not detected. Incomplete reasoning accepted as complete.
4	Omission	What is the role of temperature t in CCA decoding?	"Temperature t controls the stopping condition; as t decreases, the search ends."	**Partially Correct.** Correctly notes t decreases; omits its role shaping acceptance probabilities α_0, α_1 in the main loop.	**Incorrect.** "t does not control stopping. It modulates a probability distribution over outputs."	**Hallucinated Correction.** Partially-right claim over-penalized; actual omission (acceptance probabilities) not identified. Model contradicts a claim the student got right.
5	Omission	Which agent values the entire share Z_j?	"Agent j values Z_j, since the solid blue boxes cover all of Z_j."	**Partially Correct.** Correct answer; omits that *other* agents do not value the entire share—the contrastive point the question implies.	**Correct. N/A.**	**Verdict Inflation.** Plausible claim accepted without figure verification. Contrastive omission not detected. Most common mode for omission errors: 0/15 exact feedback match at C1.

Table 5: **[2 points]** Qualitative failure analysis (Qwen3-VL-8B, text-only condition, H3 evaluation set). Two failure modes: **Verdict Inflation** (examples 2, 3, 5)—model predicts Correct with N/A feedback, silencing the coaching pipeline; **Hallucinated Correction** (examples 1, 4)—model predicts Incorrect with confident but wrong feedback. Root cause: without the figure, the model performs plausibility checking rather than evidence-based verification. Visual context (C4) reduces both modes by grounding reasoning in figure content, explaining why feedback quality improves even as PC Recall drops.

Effect of Visual CoT on failure modes. Modification 4 (C4-CoT) forces the model to produce a `Visual Analysis` field describing key figure elements before rendering a verdict. Table 6 pairs six examples where C4 and C4-CoT diverge on the same input, illustrating two patterns. *CoT helps* (examples 1–3): the visual analysis step grounds the model in figure evidence, preventing verdict inflation—the model identifies specific values or patterns that contradict the student's claims instead of accepting them at face value. *CoT hurts* (examples 4–6): the model identifies real discrepancies during visual analysis but over-penalizes them, downgrading "Partially Correct" to "Incorrect" or, in one case, over-reading the figure and inflating to "Correct." The over-correction pattern accounts for the PC Recall drop (-4.9pp) despite the FSR improvement (-13.0pp): CoT shifts the decision boundary toward detecting errors but cannot calibrate the severity of those errors into the three-class distinction.

#	Error Type	Question (abbreviated)	GT	C4	C4-CoT	CoT Visual Analysis (excerpt)	Effect
1	Factual	How does alpha affect accuracy for different sample sizes on CLASS?	Incorrect	**Correct**	**Incorrect** ✓	"The figure clearly shows distinct, non-overlapping curves for different alpha values across all three loss functions"	Helps
2	Conceptual	Which explanation method emphasizes localized features most?	Incorrect	**Correct**	**Incorrect** ✓	"LIME shows the most focused, small-region highlighting…Parzen's maps show broader, less localized changes"	Helps
3	Omission	Role of temperature t in CCA decoding?	PC	**Correct**	**PC** ✓	"The loop uses t in acceptance probabilities α_0, α_1…student omitted this modulation role"	Helps
4	Factual	Which method predicts motion of small object most accurately?	Incorrect	**Incorrect** ✓	**Correct**	"Green arrows show motion; panel (c) ground truth…" Model over-reads figure as supporting student claim	Hurts
5	Conceptual	What is the purpose of the blank space Z_5'?	PC	**PC** ✓	**Incorrect**	"Blank space is intentionally labeled as fifth component" — model flags student's "leftover" framing as fully wrong rather than partially right	Hurts
6	Omission	Which model performs best for response selection?	PC	**PC** ✓	**Correct**	"KEHNN has highest scores in all four metrics: 0.786…" — model confirms student's claim with specific values, missing the omission	Hurts

Table 6: Qualitative comparison of C4 baseline vs. C4-CoT (Modification 4), Qwen3-VL-32B. ✓ marks the correct verdict. PC = Partially Correct. Examples 1–3: CoT prevents verdict inflation by grounding in figure evidence. Examples 4–6: CoT over-corrects or over-confirms, misclassifying the severity of the error.

2 [3 POINTS] INSIGHTS

NOTE: *This section is a WIP and will be refined.*

The verdict–feedback dissociation is a task decomposition signal, not a bug to fix. Our intrinsic metric analysis sharpens what appeared in Report 2 as a surprising empirical finding into a precise, mechanistically-grounded claim. PC Recall drops from 44.3% to 19.7% as visual input is added, while VGS rises from 35.5% to 49.4%. These measure different sub-skills that are genuinely in tension at 8B scale: visual information improves value extraction into text (VGS) but introduces noise into the three-class decision boundary, particularly for the partially correct middle class that requires the most nuanced judgment. The condition routing modification confirms this: using C1 for verdict and C4 for feedback simultaneously recovers C1's PC Recall (44.3%) and C4's feedback quality (6.0% match) at no additional inference cost. This reframes the main model proposal for Report 4: rather than searching for a single configuration or fine-tuning objective that improves both metrics jointly, Study Coach should be designed as an explicit two-stage pipeline with task routing—a lightweight verdict classifier (the Verdict-Classification Head proposed in Report 2, Section 3.4) decoupled from the feedback generator. The head would be trained directly on the three-class boundary using C1 or C2 inputs, while the feedback generator continues to use C4.

Visual chain-of-thought addresses suppression but not calibration. Modification 4 tests whether the verdict–feedback dissociation can be mitigated by forcing explicit visual reasoning before the verdict. The results are instructive: CoT reduces FSR by 13pp (from 30.6% to 17.6%), confirming that structured figure analysis prevents the model from dismissing errors as correct without evidence. But PC Recall simultaneously drops by 4.9pp, revealing that CoT introduces over-correction—the model identifies discrepancies between student claims and figure content but cannot calibrate whether those discrepancies warrant "Partially Correct" or "Incorrect." This is a different failure mode from the baseline: the original C4 model under-detects errors (high FSR), while C4-CoT over-penalizes them (low PC Recall). Together with the routing result from Modification 1, this suggests that the optimal Report 4 architecture combines three components: (1) C1-based verdict classification (highest PC Recall), (2) CoT-enhanced C4 feedback generation (lowest FSR, stable VGS), and (3) the Verdict-Classification Head to learn the three-class boundary directly rather than relying on prompt-based calibration.

The two architectural ablations together form a decision tree for Report 4. The gap between VGS (49.4% at C4) and PC Recall (19.7% at C4) has two competing explanations. The first is architectural: Qwen3-VL's bolt-on SigLIP-2 encoder feeds visual tokens into a Qwen3 backbone via a staged pipeline (merger-only Stage 0, then full training), creating a representational boundary that may affect classification more than generation. The second is scale-dependent: the capacity required to use visual input reliably for three-class verdict decisions simply exceeds what 8B parameters can provide, regardless of architecture. Modifications 2 and 3 test these explanations at two operating points: 9B early fusion (isolates architecture at matched scale) and 397B early fusion (tests whether the combination of architecture change and scale increase resolves what neither alone could). The three outcomes map directly to Report 4 strategy: if Qwen3.5-9B resolves the gap, fine-tune that backbone on SPIQA+; if only 397B resolves it, use the 27B dense variant as a practical fine-tuning target (full 397B fine-tuning is cost-prohibitive but the result identifies the correct architecture family); if neither resolves it, the task requires the Verdict-Classification Head (Report 2, Section 3.4) as an explicit supervisory signal for the verdict decision, with condition routing as the production deployment strategy regardless. The condition routing result from Modification 1 is robust to all three outcomes: it already captures C1 verdict precision and C4 feedback quality simultaneously, and will remain the best zero-cost inference strategy until fine-tuning is complete.

Mixture-of-Experts architectural modification Our architectural ablation to introduce mixture-of-experts at the LM backbone resulted in improved task performance, but did not exhibit any significant expert specialization. We think it may be due to the following reasons.

Insufficient data for specialization to emerge SPIQA dataset has QA 262,524 examples in its training split. Our finetuning used the first 8,000 QA examples from this dataset. This subset may not have been representative of the full visual input distribution. Furthermore, SPIQA dataset is limited to scientific paper images which is a constrained set of visual inputs. We think better results may have been obtaind if a) additional sampled data from SPIQA+ train was used in fine-tuning b)

the fine tuning task was done using a mixture of additional VQA and non-VQA style datasets.

The visual summary signal is too weak The router operates on mean-pooled merger output. Mean-pooled merger output is a single 4096 dimensional vector (in LM embedded space) averaged over all visual tokens. Mean pooling is typically a lossy compression. Fine-grained figure-type specific information contained in individual visual tokens but may get washed out in the mean. Using a max-pooling or using an attention-weighted sum obtained via a small cross-attention layer may result in better router discrimination.

The task output is format-dominated The structured output $verdict$ / $error_category$ / $feedback$ imposes a very strong shared template across all examples regardless of figure type. This means the LM loss gradient looks similar for a table example and a plot example, because the model has more incentive to learning something like "produce this format" rather than "reason deeply about this figure type." The routing signal from the LM loss is therefore not figure-type discriminative, so the router gets low gradient pressure to specialize even if the visual summary signal were sharper. We attempted to force router load-balancing with several techniques as described in section 1.2. But this may not have had enough effect at this training dataset scale.

2.1 Team Member Contributions to this Report

Dipti Aswath Defined the three intrinsic metrics (PC Recall, FSR, VGS) and computed all values from per-example raw outputs. Designed and evaluated Modification 1 (Condition Routing). Performed qualitative failure analysis and selected representative examples. Wrote Sections 1 (Intrinsic Metrics), Modification 1, Qualitative Analysis, Section 2 (Insights), and this contributions section.

Buddhika Kahawitage Designed and implemented Mixture-of-Lora-Experts architectural modification. Generated train, validation and test SPIQA+ splits using a newly added data generation script. Conducted the fine tuning and evaluated the results with multiple ablations. Did the write-ups on MoE modification at Section 1.2 and Section 2.

Jiashan Cui *[To be filled in by Jiashan.]*

Khubi Shah Designed and implemented Modification 4 (Visual Chain-of-Thought Prompting). Created the CoT system prompt, inference scripts (`eval_multimodal_cot.py`, `eval_cot.py`), and CoT-aware parser. Ran the full N=108 C4-CoT evaluation on Qwen3-VL-32B, computed intrinsic metrics (PC Recall, FSR, VGS), and performed qualitative comparison of C4 vs. C4-CoT (Table 6). Wrote the Modification 4 section and CoT-related additions to Insights.

References

A Appendix

NOTE: *This appendix is a WIP and will be refined.*

A.1 Intrinsic Metric Computation Details

All intrinsic metrics computed from 8B per-example outputs (H3/H4 evaluation set, N=108; 52 factual, 41 conceptual, 15 omission). Ground truth: 61 partially correct, 47 incorrect, 0 correct.

PC Recall: Count predictions of "Partially Correct" where ground truth = "Partially Correct." 61 such cases in the dataset.

FSR: Count predictions where the raw output feedback field = "N/A" or empty. These correspond exactly to Correct predictions by prompt design.

VGS: For factual error examples only, among non-suppressed feedbacks, count those matching a regex for specific numerical values (*digits followed by %*, *multi-digit integers*, or *decimal numbers*). Denominators: C1=45, C2=43, C3=36, C4=34 (remaining after suppression).

Condition Routing: Matched by (`paper_id`, `question`) key across text_only and multimodal outputs. All 108 examples matched (100% coverage). Verdict and error category from text_only; feedback from multimodal.

A.2 FEEDBACK JUDGMENT COUNTS (8B, H3 EVALUATION SET, N=108)

Error Type	Condition	Match	Partial	Unmatched	Match%
Factual	text_only	7	9	36	13.5%
Factual	caption_only	13	7	32	25.0%
Factual	vision_only	15	6	31	28.8%
Factual	multimodal	18	8	26	34.6%
Conceptual	text_only	11	16	14	26.8%
Conceptual	caption_only	21	10	10	51.2%
Conceptual	vision_only	15	10	16	36.6%
Conceptual	multimodal	24	4	13	58.5%
Omission	text_only	0	7	8	0.0%
Omission	caption_only	0	7	8	0.0%
Omission	vision_only	4	3	8	26.7%
Omission	multimodal	5	3	7	33.3%

Table 7: LLM-as-Judge feedback evaluation counts (8B, N=108).